



ROOT CAUSE UNLOCKED · CONDITION KIT

# The Neuropathy Root Cause *Kit*

*A Root Health Condition Guide*

A patient's guide to reading the lab panels behind numbness, tingling, and burning in the hands and feet.

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# The Neuropathy Root Cause Kit

*A Root Health Condition Guide*

BEFORE YOU START

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# What This Guide Is For

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You already know something is wrong. The numbness, the tingling, the burning in your feet or hands did not start for no reason, and it is not something you simply have to live with. This guide exists to help you understand what your own lab results actually mean: which panels look at which part of the problem, what a meaningful result on each one suggests, and how the pieces fit together once you can see them side by side.

This is not a treatment plan and it does not replace clinical judgment. It is the missing translation layer between a stack of lab results and a clear picture of what your nerves are actually dealing with.

Understanding your labs is the first step in the Unlock Your Neuropathy framework: it is how you move from an unexplained symptom to a specific, investigable set of drivers.

What this guide does not do is just as important as what it does. It does not tell you what dose to take, what product to buy, or what your personal protocol should be. Those decisions depend on your full history and belong in a conversation with Dr. Seay or your treating provider, not in a self-serve document. If a result concerns you, or if you are experiencing any of the warning signs in the closing section, seek clinical review rather than waiting.

For educational purposes only. This guide does not constitute medical advice, diagnosis, or treatment, and reading it does not establish a patient-provider relationship.

# Panel Overview: Which Labs, and Why, for Neuropathy Specifically

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Nerve symptoms can come from several different directions at once, which is why no single lab panel tells the whole story. Each of the panels in this guide looks at a different layer of the problem, and understanding what each one is built to see helps you understand why your provider might order more than one.

The Neural Zoomer Plus is the direct autoimmune and infection lens. It screens antibody reactivity across several neurological categories, including markers aimed squarely at peripheral neuropathy and at demyelination, the loss of the insulating sheath around nerve fibers, along with broader categories covering blood-brain barrier disruption and chronic infections. The core question this panel answers is whether your immune system is actively reacting against nerve or myelin tissue, and whether that reaction is isolated to the peripheral nerves in your hands and feet or part of a broader systemic pattern.

The Organic Acids Test, usually shortened to OAT, is the functional and metabolic lens. It measures urinary metabolites that reflect your B-vitamin status, your mitochondrial energy production, and your gut microbial activity in real time. Where the Neural Zoomer asks whether your immune system is attacking nerve tissue, the OAT asks something more basic: whether your cells currently have what they need, in terms of raw materials and usable energy, to build and maintain myelin and to keep nerve tissue running.

Standard blood chemistry, the kind drawn at any LabCorp or Quest location, fills in the metabolic-driver lens through HbA1c, fasting glucose, and fasting insulin. These three numbers, read together, are how you catch diabetic and prediabetic neuropathy, which remains the single most common cause of peripheral neuropathy worldwide. Vitamin B12 with methylmalonic acid, ordered either as its own blood draw or as part of the OAT, adds functional B12 status at the cellular level rather than just the circulating serum number, because B12 deficiency is a common and reversible cause of neuropathy and a normal serum B12 result can still coexist with a real functional deficiency.

Three additional panels round out the picture for specific circumstances. The Wheat Zoomer looks at gluten peptide reactivity and gut barrier markers, occludin and zonulin antibodies, and becomes relevant when Demyelination markers are elevated because of a documented molecular mimicry relationship between gliadin and myelin basic protein. The Heavy Metals Panel measures your body burden of neurotoxic metals, including arsenic and lead, both of which are documented causes of peripheral neuropathy entirely on their own, independent of any autoimmune or metabolic driver. And the Tickborne Diseases Panel screens for *Borrelia* species, *Bartonella*, and other tickborne and opportunistic co-infections, an infectious driver that standard two-tiered Lyme testing frequently misses; the panel's own intended use lists numbness, tingling, and Bell's palsy among the qualifying neurological symptoms.

You do not need every panel to start making sense of your symptoms. Section 7 walks through which panel to prioritize based on your own history.

# Before You Read: Four Rules for Interpreting These Results

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Read patterns, not single markers. One elevated antibody, especially a borderline one, carries less weight than several markers moving together across categories. A pattern across categories, for example a Peripheral Neuropathy marker plus a Blood-Brain Barrier marker, points toward a systemic process rather than an isolated local one, and that changes what is worth investigating next.

On the Neural Zoomer Plus, borderline results matter. The panel's reference ranges were established using a large healthy-adult population, and results are reported as In Control, Moderate Risk or borderline, or High Risk or positive. A borderline result is sub-threshold immune reactivity, not nothing, particularly when several categories are borderline at once or your symptom picture is significant.

The OAT reflects your current biochemical state, not a permanent deficit. Organic acids are urinary metabolites that shift as the underlying driver shifts. An elevated methylmalonic acid today does not mean your B12 status is permanently damaged. It means something is limiting B12 utilization right now, and the useful question is what.

Testing does not replace a clinical exam. Nerve conduction studies (NCS) and electromyography (EMG), the standard tests neurology uses to directly characterize nerve damage, answer a different question than the labs in this guide. NCS and EMG tell you what damage already exists. The panels here are aimed at why it is happening. Where red-flag symptoms are present, described in the closing section, NCS and EMG belong alongside this workup, not instead of it.



# Section 1: Neural Zoomer Plus, Peripheral Neuropathy Category

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This category targets ganglioside lipids and nerve-terminal proteins directly implicated in immune-mediated peripheral nerve disease, the category most directly relevant to numbness, tingling, and burning in the hands and feet.

Anti-GM1 targets a ganglioside lipid found in peripheral nerve and muscle-cell membranes. A positive result here is associated with multifocal motor neuropathy, present in most people who carry this diagnosis, along with Guillain-Barre syndrome and lower motor neuron syndromes. Its close relative, Anti-GM2, targets a related ganglioside lipid that also functions as a peripheral nerve antigen, and shows up in the same family of conditions: peripheral neuropathies presenting with tingling, numbness, sensation loss, burning in the feet or hands, and sharp stabbing pains, plus chronic motor neuropathy and Guillain-Barre syndrome. Two things are worth knowing if either of these came back elevated. *Campylobacter jejuni* infection is a documented trigger for Guillain-Barre syndrome through a mechanism called molecular mimicry, where the immune system's response to the bacteria cross-reacts with nerve tissue, so a recent significant gastrointestinal infection is relevant history worth mentioning to your provider. And these two antibodies carry a notably high co-occurrence with celiac disease, along with a recommendation to monitor blood sugar and HbA1c, given how common peripheral neuropathy is in that population. That is an explicit tie between this category and both the gluten and gut picture and the metabolic picture described later in this guide.

The remaining three markers in this category point toward rarer, more specific conditions that warrant direct physician discussion rather than a general nerve-support approach. Anti-Hu, also called ANNA-1, targets neuronal antigens involved in maintaining neuron identity, and is the antibody most frequently associated with paraneoplastic subacute sensory neuronopathy, a nerve condition that can arise in connection with an underlying cancer. Anti-Ri, or ANNA-2, targets neuronal nuclear antigens and is similarly linked to paraneoplastic neurological syndromes. Anti-Amphiphysin targets a membrane protein at nerve terminals involved in neurotransmitter release, and is associated with stiff-person syndrome as well as paraneoplastic neuropathy. None of these three should be read as routine nerve-support findings; a positive result on any of them belongs in front of a physician promptly.

If any marker in this category is elevated, the next question is whether it co-elevates with anything in the Demyelination, Blood-Brain Barrier, or Infections categories covered in the next two sections. Co-elevation across categories points toward a systemic neurological immune process rather than an isolated local nerve problem.

## Section 2: Neural Zoomer Plus, Demyelination Category

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Myelin is the fatty sheath that insulates your nerve fibers and lets electrical signals travel quickly from node to node, a process called saltatory conduction. When antibodies target the proteins that build or organize that sheath, signal transmission slows, which produces much of the numbness-and-tingling pattern common to neuropathy.

Anti-MAG targets myelin-associated glycoprotein, a protein involved in signaling between neurons and the glial cells that support them in both the central and peripheral nervous system. A positive result is associated with a specific, well-described condition, Anti-MAG peripheral neuropathy, marked by depressed reflexes, weakness, discriminative sensory loss, and slowed nerve conduction velocity. Anti-Neurofascin targets a protein that organizes the nodes of Ranvier, the small gaps along a nerve fiber that fast signal conduction depends on, and elevation here is associated with chronic inflammatory demyelinating polyneuropathy, Guillain-Barre syndrome, and multifocal motor neuropathy. Anti-MOG targets a minor protein sitting on the outermost surface of myelin, exposed and accessible to antibodies, and is linked to MOGAD, a demyelinating disorder distinct from multiple sclerosis whose symptoms include tingling in the arms and legs, muscle weakness, and impaired balance.

Anti-MBP targets myelin basic protein, the second most abundant protein in myelin and a marker of active demyelination when antibodies against it appear; because Anti-MBP is generally absent in healthy people, a positive result here carries real weight, and is associated with multiple sclerosis and other demyelinating disease. Anti-PLP targets a protein that makes up roughly half the protein content of adult central nervous system myelin and is essential for myelin compaction; elevation is associated with multiple sclerosis and optic neuritis, generally reflecting broader myelin autoimmunity rather than an isolated peripheral finding. Anti-Tubulin targets the structural protein of microtubules, the intracellular transport network inside nerve cells, and is associated with Guillain-Barre syndrome and chronic inflammatory demyelinating polyneuropathy; it also runs notably high in autoimmune thyroid disease, elevated in roughly half of Hashimoto's cases and close to that in Graves' disease, and mercury and other heavy metals are known to interfere with the way tubulin assembles into microtubules in the first place.

Two connections are worth carrying with you into later sections of this guide. The first is the gluten connection: myelin basic protein shares a documented molecular mimicry relationship with gliadin, the protein fraction of gluten, which is the basis for why gluten elimination shows up as a recurring theme in functional approaches to demyelinating neuropathy. If your Anti-MBP came back elevated, the Wheat Zoomer in Section 6 is the natural next test to evaluate gluten cross-reactivity. The second is the heavy-metal connection just mentioned under Anti-Tubulin: because mercury and other heavy metals can interfere with microtubule assembly, an elevated Anti-Tubulin result is a reasonable trigger to run the Heavy Metals Panel described later in this guide.

## Section 3: Systemic Context, Blood-Brain Barrier and Infections Categories

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These two categories are not the primary focus for most peripheral neuropathy cases on their own, but a co-elevation with either one changes how a Peripheral Neuropathy or Demyelination finding should be read: from an isolated local nerve issue to part of a broader systemic process.

The Blood-Brain Barrier Disruption category includes Anti-S100B, Anti-Microglia, and Anti-GFAP, which reflect glial and astrocyte disruption more generally and are less specific to peripheral nerve symptoms on their own. One marker in this category deserves specific attention for neuropathy, however: Anti-GRP78 targets a stress protein found on brain microvascular endothelial cells, and is directly associated with numbness or loss of sensation throughout the body as a symptom, along with weakness in the limbs and painful spasms. That makes it one of the few markers in this category with a documented direct link to peripheral sensory symptoms rather than only central ones.

Within the Infections category, two viral markers have the clearest peripheral-nerve mechanisms. Anti-CMV is the more directly relevant of the two: cytomegalovirus infection is strongly associated with Guillain-Barre syndrome, and a majority of CMV-associated Guillain-Barre cases carry detectable Anti-GM2 antibodies, the same antibody covered in Section 1 that binds peripheral nerves and disrupts neuromuscular transmission. That is an explicit bridge between an Infections-category finding and the Peripheral Neuropathy category. Anti-EBV seropositivity, particularly to the EBNA antigen, correlates with molecular mimicry against myelin basic protein instead, tying it to the Demyelination category rather than to peripheral nerve tissue directly.

If your Peripheral Neuropathy or Demyelination markers are elevated alongside Blood-Brain Barrier or Infections markers, that combination is worth bringing to a Root Discovery Session as a pattern, rather than treating each marker as a separate, unrelated finding. One caveat belongs here as well: tick-borne infection is not covered on the Neural Zoomer Plus. If you have a history of tick exposure, that requires the separate Tickborne Diseases Panel discussed in Section 6.

## Section 4: Organic Acids Test, Markers Relevant to Nerve Symptoms

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Methylmalonic acid, usually abbreviated MMA, is the single most directly relevant OAT marker for nerve symptoms. Vitamin B12 is a required cofactor for the enzyme that converts methylmalonyl-CoA to succinyl-CoA. When B12 is insufficient or cannot be used at the cellular level, methylmalonyl-CoA gets redirected into methylmalonic acid instead, which then shows up as elevated MMA in urine. Because this reflects functional, cellular-level B12 utilization rather than the circulating serum number, MMA can be elevated even when a standard serum B12 test looks perfectly normal. Functional B12 deficiency specifically tends to travel with diabetes, pregnancy, cancer, renal insufficiency, and conditions marked by increased oxidative stress, and it is worth knowing that metformin, a common diabetes medication, can interfere with B12 absorption and elevate MMA on its own, an additional connection between the metabolic and nutrient-depletion pictures in neuropathy.

Folate, vitamin B9, works alongside B12 in the same pathway, the methylation cycle, and the two are worth understanding together rather than as separate line items on a lab report. The methylation cycle is also responsible for clearing homocysteine, an amino acid that becomes neurotoxic and inflammatory when it builds up in the blood rather than being cleared efficiently. When either B12 or folate runs short, methylation slows, homocysteine rises, and elevated homocysteine is associated with increased inflammation, oxidative stress, and vascular changes, all of which work against healthy nerve tissue. A standard serum folate test, drawn as part of routine blood work, tells you how much folate is circulating at that moment, and it is a genuinely useful starting point. But it does not always tell the full story, because circulating folate still has to be converted, through several enzymatic steps, into its active form, 5-methyltetrahydrofolate, sometimes written as methylfolate or 5-MTHF, before your cells can actually use it. A common variant in the MTHFR gene slows that conversion step. Someone carrying this variant can post an entirely normal serum folate number while still running functionally short on usable folate at the cellular level, the same normal-on-paper-but-insufficient-in-practice pattern already described above for B12 and methylmalonic acid. This is part of why methylation-pathway markers are worth reading together rather than one at a time: homocysteine gives a downstream readout of how well the whole methylation cycle is running, and on the OAT itself, urinary uracil serves a related purpose, because folate is required to convert uracil into thymine during DNA synthesis, so elevated uracil is a recognized signal of folate deficiency or impaired methylation even when serum folate looks adequate.

The practical takeaway, at the educational level this guide operates on, is that a normal folate result does not automatically rule folate out as a contributor to nerve symptoms, particularly if homocysteine is elevated, if uracil is elevated on the OAT, or if an MTHFR variant is known or suspected. Methylfolate, often paired with methyl-B12 and P5P, the active form of B6, is a form that bypasses the conversion step an MTHFR variant slows down, and it has been studied and used in supporting people with peripheral neuropathy. What that

means for your own results, and whether it is worth pursuing, is exactly the kind of individualized decision this guide does not make; that conversation belongs in a Root Discovery Session.

Pyridoxic acid, a B6 marker, reflects a nutrient with an unusually broad job description: B6 is a cofactor for more than one hundred enzymatic reactions, including neurotransmitter synthesis and myelin formation, and it also supports B12 absorption. Both deficiency and long-term excess, generally meaning intake above 200 milligrams a day sustained over time, are documented causes of sensory neuropathy, which is why B6 status is worth tracking in either direction rather than assuming that more is automatically better.

Nerve cells, like brain cells, are highly energy-demanding, and the OAT measures several Krebs cycle intermediates, citric acid, cis-aconitic acid, alpha-ketoglutarate, succinic acid, and fumaric acid among others, that together describe how well your mitochondria are producing usable energy. A pattern of elevation across several of these markers points toward impaired mitochondrial energy production, a recognized contributor to both nerve dysfunction and the fatigue that so often accompanies it. The OAT's gut microbial markers connect back to this same picture in an unexpected way: abnormal gut microbiota is itself a contributor to elevated methylmalonic acid, through increased propionic acid production that interferes with B12-related pathways in the liver, an internal link between the gut picture and the B12 picture that is easy to miss if each marker is read in isolation. The full interpretation of the gut-specific markers is covered in the companion guide, *Unlock Your Gut*.

## Section 5: Metabolic and Nutrient Panels Beyond the Zoomer Panels

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Diabetic and prediabetic neuropathy is the most common form of peripheral neuropathy worldwide, and the specialty panels described above are not built to measure blood sugar regulation directly. For this piece of the picture, standard chemistry panels, available through any LabCorp or Quest draw, fill the gap.

HbA1c reflects your average blood sugar over roughly the past two to three months. It is a standard test, but looking at an optimal range rather than only a non-diabetic cutoff is worth understanding: chronically elevated blood sugar drives the formation of advanced glycation end products, compounds that bind receptors and trigger inflammatory signaling capable of damaging nerve tissue over time, independent of a formal diabetes diagnosis. Fasting glucose and fasting insulin, measured together, show how much insulin your body needs to keep blood sugar stable. Elevated fasting insulin alongside normal fasting glucose is a sign of developing insulin resistance, often present years before glucose itself becomes abnormal, and this pairing can catch a metabolic driver of neuropathy that a glucose-only panel misses entirely. HOMA-IR is simply a calculated insulin resistance score derived from those same two numbers, a useful way to quantify insulin resistance from the same draw rather than ordering a separate test.

At Root Health, the ranges considered optimal are deliberately tighter than the standard laboratory cutoff for normal, because the aim is to catch a developing metabolic driver of nerve damage long before it would ever meet the threshold for a diabetes diagnosis. A value drifting above the top of any of these ranges, even one still sitting inside the lab's wider normal range, is worth reading as an early metabolic signal rather than a clean result, and it is exactly the kind of finding that belongs in a conversation about your whole picture rather than a number to act on by itself.

These are clinical targets used to guide investigation. They are not validated diagnostic thresholds, and a result outside a functional window is not a diagnosis of anything.



| Marker                   | Unit   | Standard lab range | Root Health functional target |
|--------------------------|--------|--------------------|-------------------------------|
| HbA1c                    | %      | 4.0 to 5.6         | 4.6 to 5.3                    |
| Glucose, fasting         | mg/dL  | 70 to 99           | 75 to 86                      |
| Insulin, fasting         | uIU/mL | 2.0 to 10.0        | 2.6 to 5.0                    |
| HOMA-IR                  | index  | 0 to 2.5           | 0 to 1.4                      |
| Vitamin B12              | pg/mL  | 200 to 1100        | 500 to 900                    |
| Methylmalonic acid (MMA) | nmol/L | 0 to 243           | 0 to 162                      |
| Folate (RBC)             | ng/mL  | 400 to 2000        | 600 to 1500                   |
| Homocysteine             | umol/L | 0 to 10.0          | 0 to 7.0                      |
| Vitamin D (25-OH)        | ng/mL  | 20 to 100          | 40 to 80                      |
| Magnesium (RBC)          | mg/dL  | 4.0 to 7.2         | 5.2 to 6.8                    |
| Ferritin                 | ng/mL  | 25 to 300          | 50 to 150                     |
| hs-CRP                   | mg/L   | 0 to 1.0           | 0 to 0.5                      |
| TSH                      | mIU/L  | 0.4 to 4.0         | 1.0 to 2.5                    |

Two of these deserve a note. HOMA-IR has no meaningful lower bound; a low score reflects good insulin sensitivity, so only the upper edge of the window is the signal. And ferritin is the marker most likely to mislead here, because it rises with inflammation as well as with iron stores, so a ferritin sitting inside the functional window alongside an elevated hs-CRP does not reliably confirm adequate iron.

Vitamin B12 with methylmalonic acid, ordered as a standalone blood draw rather than through the OAT, uses the same functional-deficiency logic described in Section 4: ordering serum B12 alongside MMA, rather than serum B12 alone, is what catches a functional deficiency that a normal serum number would otherwise hide. Serum folate and homocysteine belong in the same conversation. Ordered together, they extend that same functional lens from B12 to folate, and a homocysteine number that is elevated despite a normal serum folate is itself a reason to ask whether an MTHFR variant or another methylation bottleneck, rather than dietary intake, is the limiting factor, exactly as described in Section 4.

## Section 6: Additional Panels Worth Knowing About

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The Wheat Zoomer becomes relevant given the documented molecular mimicry between gliadin and myelin basic protein described in Section 2. Its gluten peptide reactivity testing, along with gut barrier markers occludin and zonulin antibodies, make it a natural companion test for anyone with elevated Demyelination markers. Zonulin specifically regulates the tight junctions between intestinal cells, and dysregulated zonulin signaling is a documented biomarker for increased gut barrier permeability, sometimes called leaky gut, which has its own downstream effects on systemic inflammation.

The Heavy Metals Panel is worth considering for two metals in particular that have a directly documented, independent peripheral neuropathy mechanism, separate from any autoimmune or metabolic driver. Chronic oral exposure to inorganic arsenic is documented to cause peripheral neuropathy along with anemia and skin changes, and lead is documented to commonly cause neuropathy in adults, in addition to its better-known effects on cognition and mood. This panel earns particular consideration when nerve symptoms have not responded to nutrient, metabolic, or immune-focused work, or where an exposure history, such as older housing, well water, or occupational exposure, is present.

The Tickborne Diseases Panel fills a gap that standard testing genuinely misses: standard two-tiered Lyme testing carries a documented sensitivity of only 30 to 40 percent in early disease, meaning a negative standard screen tells you far less than most patients assume. Neurological symptoms including numbness, tingling, dizziness, and Bell's palsy are listed as qualifying reasons to run the full tickborne panel rather than a standard Lyme screen alone, and *Borrelia garinii* in particular is associated with neurological presentations of Lyme disease. *Bartonella*, one of the tick-borne co-infections tested on this panel, produces its own symptom picture, fever, fatigue, muscle and joint pain, and swollen lymph nodes, that can overlap with and complicate a Lyme diagnosis rather than standing apart from it cleanly.

There is one further pattern worth naming, with a clear caveat about where it comes from. In Dr. Seay's clinical experience, patients who turn out to have a *Bartonella* picture often describe their nerve sensations in a particular way: fizzing or electric rather than simply numb, along with buzzing, internal vibrations, or brief jolts. This is a clinical observation offered as something to mention to your provider, not a documented diagnostic sign. The published literature supports the general neurological symptom picture described in the paragraph above; it does not establish any specific sensation quality as *Bartonella*-specific. Treat it as a reason to raise the question, never as a reason to conclude the answer.



# Section 7: Testing Decision Framework

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You do not need to order everything at once, and where to start depends on your own history. If you have diabetes, prediabetes, or any history of blood sugar issues, start with HbA1c, fasting glucose, and fasting insulin from Section 5, run alongside the Neural Zoomer Plus, since the metabolic and autoimmune pictures are common enough together that checking both from the outset saves a second round of testing later. If your nerve symptoms are progressive or unexplained with no clear metabolic driver in your history, the Neural Zoomer Plus paired with the OAT is the core combination, giving you the immune and the functional-metabolic pictures side by side.

If you have a personal history of gluten sensitivity, a family history of autoimmune disease, or elevated Demyelination markers on the Neural Zoomer, add the Wheat Zoomer to evaluate the gluten connection described in Section 2. If you have a history of tick exposure, or your nerve symptoms present with prominent numbness, tingling, or a Bell's palsy-type pattern, go directly to the Tickborne Diseases Panel rather than relying on a standard Lyme screen, for the sensitivity reasons described in Section 6. If your history includes older housing, well water, or occupational exposure, the Heavy Metals Panel is worth prioritizing. And if you have a normal serum B12 result despite classic B12-deficiency symptoms, B12 with methylmalonic acid, ordered either as a standalone blood draw or as part of the OAT, is the test that can catch what the serum number alone missed.

## Section 8: Reading Your Results Together, Pattern Recognition

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Certain combinations of results are more informative read together than any one of them is alone. If your Peripheral Neuropathy category is elevated, particularly Anti-GM1 or Anti-GM2, alongside a positive Wheat Zoomer for gluten reactivity, that combination reflects a documented high co-occurrence between these two antibodies and celiac disease, and points toward the gut and immune pictures being connected rather than coincidental; Section 1 and Section 6 are the places to look next.

If Anti-MAG or Anti-MBP is elevated alongside a positive Wheat Zoomer and elevated methylmalonic acid, the picture that emerges is molecular mimicry-driven demyelination compounded by a functional B12 deficit that is limiting your body's capacity to repair myelin, a combination worth reading across Sections 2, 4, and 6 together. If Peripheral Neuropathy or Demyelination markers are elevated alongside Anti-GRP78 or Anti-CMV, that points toward a systemic neurological immune process affecting both peripheral and central nervous tissue rather than an isolated local nerve problem, following the GRP78-sensation and CMV-GM2-Guillain-Barre mechanisms described in Section 3.

If Anti-GM1 or Anti-GM2 is elevated alongside elevated HbA1c or fasting insulin, that combination is a specific, documented reason to monitor blood sugar closely, given how common peripheral neuropathy is in people carrying these antibodies; Sections 1 and 5 cover this pairing. If your Neural Zoomer Plus comes back normal or negative but bacterial or fungal OAT markers and methylmalonic acid are elevated, your nerve symptoms are more likely metabolic and gut-driven than autoimmune, which points you toward Section 4 and the companion guide, *Unlock Your Gut*, rather than further immune workup. And if you have numbness or tingling with a history of tick exposure and a negative or inconclusive standard Lyme test, remember that standard two-tiered testing misses 60 to 70 percent of early Lyme disease; a dedicated Tickborne panel, covered in Section 6, is the appropriate next step, not a repeat of the same screen that already came back inconclusive.

# When to Seek Clinical Review

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This guide is self-serve education. It explains what your labs are showing you; it does not replace an individualized clinical evaluation, and it does not give dosing, sequencing, or treatment instructions. Bring your actual results to a Root Discovery Session or your appointment with Dr. Seay for interpretation specific to your case.

Seek clinical review without delay, rather than waiting on lab results, if you experience any of the following: a fall related to numbness or balance loss, rapidly progressing weakness in addition to numbness, symptoms spreading quickly over days rather than months, or any difficulty with breathing or swallowing alongside nerve symptoms. These are red-flag presentations where a conventional nerve conduction study or electromyography should be pursued alongside, not instead of, the functional workup in this guide.

Book a free Root Discovery Session to review your results and get clarity on which panel is the right next step for your specific case: [rootcauseunlocked.com/discovery](https://rootcauseunlocked.com/discovery)

# Track What Changes, From Anywhere

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Reading your labs once tells you where you are starting. What tells you whether anything is working is what happens to your symptoms over the following weeks, and that is worth recording rather than reconstructing from memory at your next appointment.

The Root Tracker is free, works on any device, and keeps your entries on your own device. Turn on recovery tracking if nerve symptoms are limiting what you can do day to day: [rootcauseunlocked.com/tracker?utm\\_source=guide&utm\\_medium=ebook&utm\\_campaign=neuropathy-lab](https://rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=neuropathy-lab)

Dr. Seay is licensed in North Carolina, so consultations are available to North Carolina residents. The tracker and this guide are not geography-bound.

# Gaps and Assumptions Note

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This guide covers the lab panels most directly relevant to peripheral nerve symptoms.

**Resolved 2026-08-06.** The Section 5 functional ranges are now published as a table, transcribed value by value from the Root Health blood chemistry interpreter rather than paraphrased. One correction was made in the process: the earlier draft gave the HOMA-IR optimal window as 0.5 to 1.4, but the source defines it as 0 to 1.4 with no lower bound. The invented 0.5 floor is removed.

**Still open for Dr. Seay.** The Bartonella sensation-quality pattern in Section 6 is now written explicitly as clinical observation rather than a documented sign, since no published source establishes a Bartonella-specific sensation quality. Confirm whether that framing matches your clinical experience, or cut it.

Book a complimentary Root Discovery Session to review your results: [rootcauseunlocked.com/discovery?utm\\_source=guide&utm\\_medium=ebook&utm\\_campaign=neuropathy-lab](https://rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=neuropathy-lab)

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For educational purposes only. This guide does not constitute medical advice or establish a patient-provider relationship.

# Read this part first

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This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient. Nothing here is a reason to change a medication on your own, and one section specifically warns against a change people make without telling their prescriber.

If you have rapidly progressing weakness, symptoms spreading over days rather than months, a fall related to numbness or balance, or any difficulty breathing or swallowing alongside nerve symptoms, seek care now rather than starting a diet.

# What you are actually working toward

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Most diet advice for neuropathy arrives as a list of anti-inflammatory foods, which is fine as far as it goes and tells you almost nothing about what to expect. You end up eating more salmon and blueberries, feeling vaguely virtuous, and six months later you still cannot tell whether any of it helped.

This guide is built the other way around. It starts with the one dietary change that has genuinely high-quality trial evidence behind it, tells you honestly where that evidence is strong and where it has been oversold, and then gives you a way to test the rest on yourself so you end up with answers instead of habits.

# What the evidence actually shows, including the part usually left out

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**Blood sugar control is the intervention with real evidence, and the honest version has a catch worth knowing.**

A Cochrane review examined enhanced glucose control for preventing and treating diabetic neuropathy.<sup>1</sup> In **type 1 diabetes**, pooling 1228 participants, tighter control significantly reduced the risk of developing clinical neuropathy, with an annualized risk difference of 1.84%. The review rates this as high-quality evidence.

In **type 2 diabetes**, pooling 6669 participants across four studies, the annualized risk difference was 0.58% and **it did not reach statistical significance**, at  $P = 0.06$ . Nerve conduction and vibration threshold measures did improve significantly in both groups.

This matters because most people with diabetic neuropathy have type 2, and the confident claim that tight glucose control prevents neuropathy is largely a type 1 finding being applied to a type 2 audience. The direction of effect in type 2 is favorable and the nerve measures do improve. It simply has not been demonstrated for clinical neuropathy at the level people assume.

**And there is a real cost on the other side.** The same review found that enhanced glucose control significantly increased severe adverse events including hypoglycemic episodes, in both types, and states plainly that this needs to be weighed when judging the risk and benefit.<sup>1</sup> Driving your numbers down aggressively is not a free action, and it is a decision to make with your prescriber rather than alone with a continuous glucose monitor and this guide.

**What about the Mediterranean diet?** A 2025 systematic review and meta-analysis of 14 studies found that adherence was associated with reduced risk of diabetic retinopathy and diabetic nephropathy. For **neuropathy specifically, the results were inconclusive due to study heterogeneity**, and the authors say so directly.<sup>2</sup> So a Mediterranean pattern is a defensible way to eat for several good reasons. Preventing neuropathy is not yet one of the proven ones, and anyone telling you otherwise is reading past the paper.

That is the honest state of the evidence. What follows is built on it rather than around it.



# Part one: blood sugar, the biggest lever you have

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Even with the caveat above, this is where the effort belongs, because it is the only dietary target with high-quality trial evidence behind it in any diabetic population, and because the mechanism is not in dispute.

**The practical target is not a number, it is the size of the swing.** Two people can have the same HbA1c, one running steadily and one spiking after every meal and crashing between. Post-meal spikes are the part a fasting number does not show you.

**Four changes that flatten the curve, in order of how much they usually help:**

**Eat protein and fat before carbohydrate at the same meal.** Same food, same portions, different order. This is the lowest-effort change in the guide and it requires you to buy nothing.

**Anchor every meal with protein.** Palm-sized portion, every time. This is the single most reliable way to reduce the size of a post-meal rise without counting anything.

**Move for ten minutes after eating.** A walk after your largest meal blunts the post-meal rise. It does not need to be exercise, it needs to be movement.

**Cut liquid sugar entirely, first.** Soda, juice, sweetened coffee drinks, and sports drinks hit faster and higher than any solid food. If you do one thing on this page, this is the one that changes the numbers fastest.

**What not to do:** do not start skipping meals or fasting aggressively to bring numbers down, particularly if you take insulin or a sulfonylurea. That is the fastest route to the severe hypoglycemia the Cochrane review flagged, and hypoglycemia has its own consequences.

## Part two: alcohol, which is a cause and not a lifestyle nag

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This section is short and it is not optional reading.

**Alcohol causes peripheral neuropathy directly.** A systematic review and meta-analysis of alcohol-related peripheral neuropathy examined prevalence, character, risk factors, and management across the literature.<sup>3</sup> This is a recognized clinical entity with a well-described pattern: axonal degeneration with reduced density of both small and large nerve fibers, producing decreased sensation, heightened pain response, and weakness.<sup>4</sup>

Two mechanisms run in parallel. Alcohol is directly toxic to nerve tissue, and chronic use depletes thiamine, which nerves require.<sup>5</sup>

**What this means for you.** If you drink and you have unexplained neuropathy, alcohol is on the list of causes, not the list of aggravating factors. It is worth an honest accounting of actual intake with your clinician, because it changes what should be investigated and because it is one of the few causes where stopping changes the trajectory.

I am not going to give you a number of drinks that is safe with an existing neuropathy, because I do not have evidence for one.

## Part three: gluten, where the connection is real and narrower than the internet suggests

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**Gluten neuropathy is a documented entity, not a wellness invention.** It has its own literature, its own antibody profile, and its own HLA associations. A study of the electrophysiological progression of gluten neuropathy found that most patients presented with a length-dependent neuropathy, while a subset had sensory ganglionopathy, and that specific HLA types tracked with which pattern people got.<sup>6</sup> Autonomic involvement is common too: in a separate study, two thirds of gluten neuropathy patients had abnormal sudomotor function in the hands or feet.<sup>7</sup>

**But that is a specific diagnosis, not a general reason to avoid bread.** The claim “gluten causes neuropathy” is different from “there is a form of neuropathy driven by gluten sensitivity in susceptible people.” The second is true. The first is not.

**Get tested before you eliminate, not after.** This is the single most common self-inflicted mistake in this area. Celiac testing depends on you currently eating gluten. Go gluten free for three months first and the testing becomes unreliable, and you have permanently lost the chance to get a clean answer without a deliberate gluten challenge, which is unpleasant.

So: if gluten is a serious question for you, ask for testing **while still eating gluten**, then eliminate. Not the other way around.

# Part four: the foundation pattern

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With the evidence-backed parts covered, this is the sensible background pattern. It is not a cure and I am not going to present it as one.

## **Build meals from:**

Non-starchy vegetables, in volume and variety. Leafy greens, cruciferous vegetables, and anything with color.

Protein at every meal. Fish, poultry, eggs, meat, or legumes if they agree with you.

Fats from whole sources. Olive oil, avocado, nuts, and seeds. Fatty fish twice a week gives you EPA and DHA from food rather than a capsule.

Low-glycemic fruit. Berries especially, because they carry the least sugar for the most volume.

Herbs and spices generously. Turmeric, ginger, cinnamon, garlic, rosemary. Cheap, pleasant, and they cost you nothing to include.

## **Reduce:**

Liquid sugar, covered above and worth repeating.

Ultra-processed food, which is the practical way to reduce refined carbohydrate, industrial oils, and additives all at once without auditing labels.

Alcohol, covered above.

**A note on some common advice I am leaving out.** You will see a rule that you should drink half your body weight in ounces of water daily. I cannot find evidence for that specific number, so it is not in this guide. Drink enough that you are not thirsty and your urine is pale. If you take a diuretic or have kidney or heart disease, fluid targets are a medical question, not a wellness one.

# Part four and a half: absorption, which is where diet quietly fails

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You can eat a nutrient and still not get it. For nerve symptoms this matters more than it does for most conditions, because the nutrients nerves depend on most are also the ones most easily blocked.

**Vitamin B12 is the main one, and several very common medications interfere with it.**

Proton pump inhibitors interfere with B12 absorption from food by slowing the release of gastric acid, and can lead to deficiency. H2 blockers work the same way, reducing the acid needed to separate B12 from food protein. Metformin might reduce B12 absorption and significantly lower serum concentrations.

None of that is a reason to stop any of those medications, and I want to be unambiguous about that. It is a reason to have B12 checked properly rather than assumed, and the lab guide in this kit explains why serum B12 alone can look fine while the cell is short.

**Gut surgery changes the arithmetic entirely.** After gastric bypass or sleeve gastrectomy the anatomy that absorbs B12 is altered, so dietary intake stops being a reliable route.<sup>8</sup> If you have had bariatric surgery and you have nerve symptoms, this belongs near the top of the investigation, not the bottom.

**One trap almost nobody knows about: zinc causes copper deficiency, and copper deficiency causes a neuropathy that looks exactly like B12 deficiency.**

Copper deficiency polyneuropathy is described as virtually indistinguishable from the nerve damage caused by low B12.<sup>9</sup> And excess zinc is a recognized driver of it, including after bariatric surgery and from sustained zinc supplementation.<sup>10</sup> Zinc is in a great many immune-support products and people take it for months without thinking of it as a drug.

This is case-report level evidence rather than trial evidence, so treat it as a thing worth knowing rather than a common cause. But if you take zinc regularly and have unexplained neuropathy, it is a cheap question to ask and an easy one to miss.

**What to do about all of this:** none of it is solved by eating differently. It is solved by testing. That is not a dodge, it is the honest boundary of what a diet guide can do for you.

# Part four and three quarters: eating for sleep, because sleep sets your pain threshold

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This section is in a diet guide because the strongest lever food has on nerve pain in the short term may be indirect.

**Sleep loss measurably lowers your pain threshold.** In a study characterizing the brain mechanisms, acute sleep deprivation amplified pain reactivity in the primary somatosensory cortex while blunting activity in the regions that evaluate and contextualize pain. It expanded the temperature range at which a stimulus was classified as painful, specifically by lowering thresholds. And the researchers found that ordinary night-to-night changes in sleep within a single person predicted day-to-day changes in experienced pain.<sup>11</sup>

That last part is the useful one. This is not just about severe insomnia. A normal bad night makes tomorrow hurt more.

**Eating changes that protect sleep, in order of impact:**

**Stop drinking alcohol to fall asleep.** It shortens the time to fall asleep and degrades the second half of the night. Given part two, this one is doing double duty.

**Move caffeine earlier.** Caffeine has a long half-life and afternoon coffee is still working at bedtime for many people. Try a hard stop at noon for two weeks before deciding it does not affect you.

**Do not go to bed hungry or overfull.** Both fragment sleep. A small protein-containing snack is better than either extreme if dinner was early.

**Keep the last large meal away from bedtime.** Reflux and blood sugar swings both worsen when you lie down soon after eating.

If nerve pain is what wakes you, fixing sleep hygiene will not fix that on its own, and the honest answer is that pain waking you is a treatment conversation, not a diet one.

# Part five: the specific experiments worth running

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The foundation pattern is background. These are the actual tests, and each one gives you a yes or a no.

**Experiment one: liquid sugar, two weeks.** Remove every sugar-sweetened drink including juice. Two weeks is enough because the effect on post-meal glucose is immediate. Watch afternoon energy and night-time symptom intensity.

**Experiment two: meal sequencing, two weeks.** Protein and vegetables first, starch last, same meals otherwise. If you use a glucose monitor, this one is directly measurable. If you do not, watch the two hours after your largest meal.

**Experiment three: alcohol, four weeks off.** Genuinely off, not reduced. Four weeks is a reasonable read on symptom intensity and sleep quality, both of which affect how much pain registers.

**Experiment four: an elimination and reintroduction, six weeks total, and only if the first three did not settle it.** Remove the common triggers for three to four weeks, then reintroduce one food group every two to three days while tracking symptoms. Gluten, dairy, soy, corn, eggs, and nightshades are the usual candidates.

The reason this guide uses elimination and reintroduction rather than a blood test is that it directly measures your own response, which is the thing you actually want to know. Blood panels marketed as food sensitivity tests are contested, and this guide does not rely on them. Do the celiac testing question first, per part three, because that one is a real diagnosis with real testing.

**One at a time.** Running three of these simultaneously produces a feeling, not a finding.

# Part six: a week you can actually cook

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Not a meal plan to follow forever. A worked example of what the pattern looks like when it stops being abstract.

**Breakfasts, rotate any three** - Eggs with spinach and half an avocado - Plain Greek yogurt, berries, walnuts, cinnamon - Smoked salmon, sliced tomato, olive oil

**Lunches** - Big salad with a tin of sardines or leftover chicken, olive oil and lemon - Lentil soup with a side of greens - Leftovers from the night before, which is what actually happens

**Dinners** - Salmon, roasted broccoli, small sweet potato eaten last - Chicken thighs, cauliflower, big green salad - Beef and vegetable stir fry over cabbage instead of rice - Bean and vegetable chili - Whatever your household eats, with the protein and vegetables served first and the starch portion halved

**Snacks if you need them** - A handful of nuts - Apple with nut butter - Hard boiled eggs

**The pattern underneath:** protein anchors every meal, vegetables take the most space, starch appears last and in a smaller portion than habit suggests, and nothing arrives in liquid sugar form.



# Running a fair diet experiment

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**Write down where you start.** Before changing anything, record your symptoms now: worst intensity, how often, what they stop you doing, and how you slept. Diet changes are gradual and memory is unreliable about gradual change in both directions.

**One change at a time.** Yes, it is slower. It is also the only way to end up knowing something you can act on.

**Give it the stated window.** Two weeks for the blood sugar experiments, four for alcohol, six for a full elimination and reintroduction.

**Decide in advance what counts as a result.** “Maybe a bit better” is not a result. Sleeping through instead of waking twice is a result. Half as much burning by evening is a result.

**Keep what worked, drop what did not.** A restriction that produced nothing is a restriction you should stop, and this is the step almost everyone skips. People accumulate food rules the way they accumulate supplement bottles.

**Track it:** [rootcauseunlocked.com/tracker?](https://rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=neuropathy-diet)

[utm\\_source=guide&utm\\_medium=ebook&utm\\_campaign=neuropathy-diet](https://rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=neuropathy-diet)

Turn on recovery tracking if nerve symptoms limit what you can do day to day. Free, works anywhere, and your entries stay on your device.

# Realistic expectations

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Here is what the evidence in this guide actually supports.

If you have type 1 diabetes, better glucose control has high-quality evidence for reducing your risk of developing clinical neuropathy, with an annualized risk difference of about 1.84%.<sup>1</sup> That is real and it is also a modest annual number that compounds over years rather than transforming a month.

If you have type 2 diabetes, better control improves nerve conduction and vibration threshold measures, while the reduction in clinical neuropathy did not reach statistical significance.<sup>1</sup> Worth doing. Not a guarantee, and not something to pursue so aggressively that you trade nerve symptoms for hypoglycemia.

If alcohol is part of your picture, stopping addresses a cause rather than a symptom, and that is a different category of change from anything else in this guide.<sup>3</sup>

If you turn out to have gluten neuropathy, dietary change is the treatment, which is why testing before eliminating matters so much.<sup>6</sup>

For everyone else, an anti-inflammatory pattern is a reasonable way to eat with modest and largely indirect support for nerve symptoms specifically. I would rather tell you that than sell you a food list as a cure.

**Nothing in this guide reverses established nerve damage.** What diet does is remove ongoing drivers so that whatever capacity for repair you have is not fighting uphill.

# When to seek professional care

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**Immediately:** rapidly progressing weakness, symptoms spreading over days, a fall related to numbness or balance, or difficulty breathing or swallowing.

**Before eliminating gluten:** if celiac disease is a serious question, get tested while still eating gluten.

**Before aggressive glucose lowering:** if you take insulin or a sulfonylurea, talk to your prescriber about targets rather than setting your own.

**At your next appointment:** if you drink regularly and have unexplained neuropathy, or if you have been gluten free for months and never had celiac testing.

# Working with Root Health

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**Book a free Root Discovery Session:** [rootcauseunlocked.com/discovery?  
utm\\_source=guide&utm\\_medium=ebook&utm\\_campaign=neuropathy-diet](https://rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=neuropathy-diet)

Consultations are available to people located in North Carolina, where I am licensed. If you are elsewhere, the testing questions in part three and the glucose target conversation in part one work just as well in your own clinician's office.

**Not ready for a conversation yet? Start tracking:** [rootcauseunlocked.com/tracker?  
utm\\_source=guide&utm\\_medium=ebook&utm\\_campaign=neuropathy-diet](https://rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=neuropathy-diet)

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Vitamin B12 medication interaction figures in part four and a half: National Institutes of Health, Office of Dietary Supplements, Vitamin B12 Fact Sheet for Health Professionals.

For educational purposes only. This guide does not constitute medical advice, diagnosis, or treatment, and reading it does not establish a patient-provider relationship.

Root Health | Dr. Adam Seay, DC | Garner, NC | [rootcauseunlocked.com/unlock-your-neuropathy](https://rootcauseunlocked.com/unlock-your-neuropathy)

# Read this part first

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This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient. Nothing here is a reason to stop, start, or change a prescription on your own.

Two things make this guide different from most supplement lists written for nerve pain.

**It tells you what did not work.** Several products sold specifically for neuropathy have been tested properly and did not beat placebo. Those are in here by name, in their own section, with what the trial found. A list that only contains good news is not a list, it is an advertisement.

**It puts the safety section before the shopping section.** One of the most common ingredients in nerve-support formulas can cause the exact symptoms you are trying to treat. That is not a footnote, so it is near the front.

If you have rapidly progressing weakness, symptoms spreading over days rather than months, a fall related to numbness or balance, or any difficulty breathing or swallowing alongside nerve symptoms, stop reading and seek care now. Those presentations need evaluation, not a supplement plan.

# Why nutrient evidence looks thinner than it is

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**A large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back.** You cannot patent magnesium, an herb, or a food. So the evidence hierarchy this guide grades against is not neutral: it systematically over-documents patentable drugs and under-documents nearly everything else. That is measurable, not rhetorical. A Cochrane methodology review found that manufacturer sponsorship of drug and device studies produces more favourable results and conclusions than independent funding, through a bias that standard quality checks do not catch (PMID 28207928).

Two honest consequences follow, and they pull in opposite directions.

**First, “no evidence” has three different meanings, and this guide tries to always tell you which one applies.** *Tested and failed:* a real trial ran and the answer was no. *Never properly tested:* nobody could afford the trial, so absence of evidence is not evidence of absence. *Tested small and won:* real but modest trials, usually measuring markers rather than outcomes, because markers are what a small budget can afford.

**Second, the asymmetry earns nutrients humility, not automatic credit.** When isolated nutrients have received large publicly funded trials, the results have been mixed: some genuine wins, some nulls, and some harms. Small nutrient trials also carry their own publication bias toward positive results, just as drug trials tilt toward their sponsor. So this guide applies one rule in both directions: name the funding, name the evidence tier, and let you see both.

# What you are actually buying back

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Most people arriving at a guide like this have already spent money. A bottle recommended in a forum, a nerve-support blend with a long ingredient list, a second bottle when the first one did nothing. The spending is rarely the real cost. The real cost is the months that went by while you waited to see whether it was working, with no way to tell.

What this guide is meant to buy back is that time. It tells you which handful of things have real trial evidence behind them, what size effect to expect, how long to give each one before you decide, and which ones to skip so you are not running six experiments at once and learning nothing from any of them.



# Supplements are supportive, not the main event

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The single most effective thing for the most common form of neuropathy is not in a bottle. Where nerve symptoms are driven by blood sugar, glycemic control does more than any supplement in this guide. Where they are driven by a B12 deficiency, correcting the deficiency is the treatment and everything else is decoration. Where they are driven by a medication, a toxin, or a chronic infection, the answer is upstream of the supplement aisle entirely.

Supplements earn their place as support around a cause that is being addressed. They are a poor substitute for finding the cause, which is what the lab guide in this kit is for.

# How to read this guide

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Everything is sorted into three tiers.

**Foundation** is the short list with the best evidence and the widest applicability. If you only do one thing, do something from here.

**Targeted** depends on what is actually driving your symptoms. These are not for everyone, and picking from here without knowing your driver is guessing.

**Ask your practitioner first** contains no products and no links. It is a list of things worth raising with a clinician, with the specific questions to ask and what they will want to know before answering. Some things genuinely should not be bought off a page, and pretending otherwise would make the rest of this guide less trustworthy.

Every entry gives you what it is, what the evidence actually shows including where it is weak, a commonly studied amount, the form that was studied, and the cautions that matter.

# Before you start: the medication conversation

This population is heavily medicated. Metformin, gabapentin, pregabalin, duloxetine, amitriptyline, SSRIs, tramadol, blood thinners, and proton pump inhibitors all show up routinely in people with nerve symptoms, and several of them interact with things in this guide.

Find your row before you buy anything.

| If you                                                | Avoid, cap, or watch                                                              | Do this instead                                                                                                                                      |
|-------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Take metformin or a proton pump inhibitor             | Assuming your B12 is fine because it was checked once                             | Ask for B12 <b>with methylmalonic acid</b> , since serum B12 alone can look normal while the cell is short                                           |
| Take an SSRI, SNRI, duloxetine, or tramadol           | Serotonin-active products including 5-HTP, SAMe, St John's wort, and L-tryptophan | Ask before adding any of them, and learn the stop-and-call symptoms below                                                                            |
| Take warfarin or another anticoagulant                | High-dose vitamin E                                                               | Tell your prescriber about omega-3 rather than avoiding it. The bleeding warning is weaker than its reputation, but periodic INR checks are sensible |
| Are in or recently finished chemotherapy              | <b>Acetyl-L-carnitine</b>                                                         | A randomized trial found it made chemotherapy-related neuropathy worse, with the effect still present two years later                                |
| Take insulin or another glucose-lowering drug         | Unmonitored alpha-lipoic acid                                                     | ALA can lower blood glucose. Monitor more closely when starting, and know your hypoglycemia symptoms                                                 |
| Already take a B-complex or a multivitamin            | Adding a second product containing B6                                             | Add up B6 from <b>all</b> sources and keep the total under the ceiling described below                                                               |
| Have had gastric bypass, sleeve, or other gut surgery | Assuming an oral B12 will absorb                                                  | Ask about absorption testing and non-oral routes                                                                                                     |
| Take levothyroxine                                    | Taking minerals at the same time of day                                           | Separate mineral supplements from thyroid medication by several hours                                                                                |
| Are pregnant, nursing, or trying to conceive          | Everything in this guide by default                                               | Ask first. Almost nothing here was studied in pregnancy                                                                                              |

**Stop and call your prescriber** if you develop agitation, confusion, a racing heart, shivering, muscle twitching or rigidity, heavy sweating, diarrhea, or fever after adding a serotonin-active supplement to an antidepressant. That combination is the top safety concern in this category.

# The vitamin B6 problem, which most neuropathy formulas get wrong

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This deserves its own section because it is the one place where a supplement marketed for nerve pain can cause nerve pain.

**High vitamin B6 intake causes a sensory neuropathy.** A 2023 systematic review in *Nutrients* reviewed twenty articles and concluded that higher B6 levels, which usually follow from taking supplements, can produce a predominantly and sometimes exclusively sensory neuropathy of the axonal type.<sup>1</sup> The symptoms are numbness, tingling, and burning in the hands and feet. If that sounds exactly like what you already have, that is the problem. It is difficult to tell B6 toxicity apart from the neuropathy you came in with.

**The ceiling is 100 mg per day from all sources combined.** That is the Tolerable Upper Intake Level for adults. Severe progressive sensory neuropathy with loss of coordination has been described with chronic intake of 1 to 6 grams daily over 12 to 40 months, and sensory neuropathy has been reported at doses below 500 mg per day, which is why the observed safe dose was halved to set the 100 mg limit.

**Now the reassuring half, because the scary half alone is misleading.** The same systematic review found that B6 supplementation, even as part of a nutritional multivitamin, has not been shown to be harmful at permitted daily doses in people who already have peripheral neuropathy.<sup>1</sup> A separate study of 261 patients with chronic idiopathic axonal polyneuropathy found that elevated plasma B6 levels were not related to neuropathy severity, nerve conduction results, or examination findings.<sup>2</sup> Nobody develops this from a standard multivitamin.

**So the real risk is stacking, not any single product.** A B-complex, plus a multivitamin, plus a nerve-support formula, each with a label that looks perfectly reasonable on its own, can quietly clear 100 mg between them. Nerve formulas in particular often carry high B6 precisely because it is marketed as a nerve vitamin.

**What to do:** add up the B6 in everything you take, including anything labeled nerve support, energy, or stress formula. Keep the total under 100 mg per day. If you are already above it, that is worth a conversation, because B6 neuropathy generally improves after stopping when it is caught early.

**And do not take B6 expecting it to treat your neuropathy.** The same review found no firm evidence that low B6 causes peripheral neuropathy, and no data at all on B6 as a standalone treatment, only as part of combinations.<sup>1</sup>

## Tier 1: Foundation

# Alpha-lipoic acid

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**What it is.** An antioxidant made in the body that is soluble in both water and fat, which is unusual and lets it act in more tissue compartments than most antioxidants.

**What the evidence actually shows.** This is the best-studied supplement in the whole category, and the picture is genuinely good in the short term and genuinely unproven in the long term. Both halves matter.

In the SYDNEY 2 trial, 5 weeks of oral ALA at 600, 1200, or 1800 mg daily reduced total symptom score by 51%, 48%, and 52% respectively, compared with 32% on placebo, all statistically significant.<sup>3</sup> The proportion of people achieving at least a 50% reduction in symptoms was 62%, 50%, and 56% across the three doses versus 26% on placebo. Stabbing pain, burning pain, and patients' own global assessment all improved.

**The most useful finding in that trial is about dose.** 600 mg performed as well as 1800 mg, and side effects rose with dose. Nausea, vomiting, and dizziness were dose-dependent.<sup>3</sup> There is no reason to buy the megadose, and a good reason not to.

A systematic review pooling oral trials found a total symptom score reduction of 1.95 points versus placebo at 600 mg oral.<sup>4</sup> A critical appraisal covering 1160 participants across the ALADIN, SYDNEY, ORPIL, SYDNEY 2, and ALADIN III trials concluded that 600 mg orally for up to 5 weeks offers benefit in symptoms and signs without significant side effects.<sup>5</sup>

**Where it falls down, stated plainly.** The NATHAN 1 trial ran 4 years and **missed its primary endpoint.**<sup>6</sup> There was no significant difference between ALA and placebo on the primary composite measure. Some secondary measures did favor ALA, including neuropathy impairment scores, and fewer people on ALA progressed. But serious adverse event rates were higher on ALA. So the honest summary is that ALA has good evidence for reducing symptoms over weeks, and has not demonstrated that it changes the course of the disease over years.

**Commonly studied amount:** 600 mg daily, oral. **Form studied:** oral alpha-lipoic acid. Most trial evidence is on the racemic form, not R-lipoic acid, despite R-lipoic being marketed as superior. **Take it:** on an empty stomach, since food reduces absorption. **Cautions:** can lower blood glucose, which matters if you take insulin or another glucose-lowering medication. Nausea and dizziness rise with dose, which is another argument for 600 mg. Discuss with your prescriber if you have thyroid disease.

# Vitamin B12, but only if you are actually low

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**What it is.** A vitamin required to build and maintain myelin, the insulating sheath around nerve fibers.

**What the evidence actually shows.** B12 deficiency is one of the few genuinely reversible causes of peripheral neuropathy, which is exactly why it belongs at the foundation. But the value is in correcting a real deficiency, not in taking B12 because you have nerve symptoms.

For B vitamins as a treatment in people who are not deficient, a 2026 review and meta-analysis of 13 randomized trials covering 834 people with diabetic peripheral neuropathy found a mixed picture.<sup>7</sup> Neuropathy screening questionnaire scores improved. Examination scores improved statistically but **did not reach the minimum clinically important difference**. Pain measured on standard numeric or visual analogue scales showed no significant effect at all. Sural nerve conduction velocity and amplitude improved. Peroneal and tibial measures were inconsistent, and tibial velocity actually favored the control group. The authors concluded that heterogeneity and risk of bias preclude firm recommendations for routine use.

So: correct a deficiency, yes. Expect B vitamins to treat neuropathy in someone who is replete, no.

**Get tested properly.** Serum B12 alone can sit inside the reference range while the cell is functionally short. Methylmalonic acid rises when B12 is inadequate at the tissue level, which is why it is ordered alongside. The lab guide in this kit covers what the numbers mean.

**Who is at real risk:** anyone on metformin or a proton pump inhibitor long term, anyone following a vegan or strict vegetarian diet, anyone with pernicious anemia, and anyone who has had gut surgery.

**Commonly studied amount:** varies enormously with the cause. Dietary shortfall, malabsorption, and pernicious anemia are treated very differently, and the last of those often needs a non-oral route. **Form:** methylcobalamin and hydroxocobalamin are the forms usually used in functional practice. **Cautions:** correcting B12 without also assessing folate can mask a folate problem. This is a test-first nutrient, not a guess-first one.

# Vitamin D, if your level is low

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**What it is.** A hormone more than a vitamin, with receptors throughout the nervous system.

**What the evidence actually shows.** A 2025 systematic review and meta-analysis of randomized trials found a significant short-term reduction in pain in painful diabetic neuropathy, with a standardized mean difference of 0.85, roughly equivalent to 11 points on common pain scales.<sup>8</sup> That is a meaningful effect size.

The honest caveats are substantial and the authors state them. Only four trials totalling 320 people were pooled. Heterogeneity was moderate. **No study reported medium or long-term outcomes at all.** Study quality varied, with concerns about allocation concealment and selective outcome reporting. Their own conclusion is that larger and more rigorous trials with longer follow-up are needed before routine vitamin D testing or supplementation can be recommended in clinical practice for this indication.

So this is promising short-term evidence, not settled practice. Correcting a documented deficiency is defensible on general grounds regardless.

**Commonly studied amount:** varies by trial and by starting level, which is why this is a test-then-dose nutrient. **Form:** D3, cholecalciferol. **Cautions:** fat soluble and genuinely possible to overdo. Retest rather than escalating indefinitely.

## Tier 2: Targeted, depending on what is driving it



# Magnesium

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**What it is.** A mineral involved in hundreds of enzymatic reactions, including nerve signal transmission and muscle relaxation.

**What the evidence actually shows.** Weaker than the tier 1 entries, and I want to be clear about that. The rationale is mechanistic and the clinical experience is favorable, particularly for the cramping, twitching, and restless legs that often accompany nerve symptoms, but there is not a body of randomized trial evidence in peripheral neuropathy comparable to what exists for ALA. It is included because it is inexpensive, low risk, and addresses a symptom cluster that genuinely bothers people, not because the trial evidence is strong.

**Form:** glycinate and malate are better tolerated than oxide, which is poorly absorbed and mostly acts as a laxative.

**Cautions:** reduce or avoid in significant kidney disease. Separate from thyroid medication and certain antibiotics by several hours. Loose stools are the usual sign of too much.

# Omega-3 fatty acids

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**What it is.** EPA and DHA, incorporated into cell membranes and involved in resolving inflammation.

**What the evidence actually shows.** Reasonable mechanistic grounding and general anti-inflammatory evidence, with limited neuropathy-specific randomized data. Treat this as supportive rather than targeted therapy, and be skeptical of anyone presenting it as a nerve treatment.

**Form:** triglyceride or re-esterified triglyceride form. **Cautions, and the usual warning here is overstated.** Fish oil is routinely described as a bleeding risk with warfarin. The actual evidence is more reassuring than the reputation: most research finds that 3 to 6 grams per day does not meaningfully affect anticoagulant status in people on warfarin, and a 2014 review concluded omega-3s do not increase clinically significant bleeding. FDA-approved omega-3 product labels say studies have not produced clinically significant bleeding episodes. Those same labels still recommend periodic INR monitoring, which is the sensible middle position: tell your prescriber you are taking it, and let them decide whether to check, rather than avoiding it out of a warning that the data does not support.

## Tier 3: Ask your practitioner first

**Nothing in this tier has a purchase link, on purpose.**

These are things where the right answer depends on your history, your medication list, and often on a test result. Selling them from a page to someone I have never assessed would contradict everything else in this guide. What follows is what to ask and what your clinician will want to know.

**Anything serotonin-active, if you take an antidepressant.** 5-HTP, SAME, St John's wort, and L-tryptophan. Duloxetine and amitriptyline are both commonly prescribed for neuropathic pain, so this overlap is not hypothetical, it is the norm. *Ask:* "I take [medication]. Is there a serotonin syndrome risk if I add this, and what symptoms should make me stop and call you?" *They will want to know:* every prescription, over-the-counter product, and supplement you take, including anything in a blend.

**High-dose B vitamins of any kind.** *Ask:* "Can we total up my B6 across everything I am taking, and is it under 100 mg?" *They will want to know:* every label, not just product names. B6 hides in blends.

**Alpha-lipoic acid, if you take glucose-lowering medication.** *Ask:* "How should I monitor my blood sugar for the first few weeks after starting this?" *They will want to know:* your current regimen and your recent readings or continuous monitor data.

**Anything at all, if you take an anticoagulant.** *Ask:* "Which of these affect bleeding risk, and do I need to stop anything before my procedure?" *They will want to know:* your indication, your target range if applicable, and your procedure schedule.

**Chelation or any product marketed for heavy metal detox.** *Ask:* "Do my actual test results support this, and what are the risks of doing it without them?" *They will want to know:* whether you have had genuine, unprovoked testing. Provoked or challenge urine metal testing is not a valid way to establish a body burden, and treatment decisions made on it are built on nothing.

**If nobody is quarterbacking your case, that is the finding.** The most common problem I see is not the wrong supplement. It is six providers each managing one piece and nobody looking at the whole list. If reading this tier made you realize you

have no one to ask, that is worth a conversation.

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There is no obligation to continue, and if the answer is that your case belongs with someone else, I will tell you that.

# What I deliberately left out, and why

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This section exists because a supplement guide that only lists winners is not being honest with you.

**Benfotiamine.** This is the one that will surprise people, because benfotiamine is in a great many nerve-support formulas and is specifically marketed for neuropathy. The BOND study randomized people with type 2 diabetes and symptomatic polyneuropathy to benfotiamine 300 mg twice daily or placebo for **12 months**, and found **no significant effect** on corneal nerve fiber length, on any secondary morphometric measure, on skin biopsy parameters, on nerve conduction, on quantitative sensory testing, on autonomic function, on sudomotor function, or on quality of life.<sup>11</sup> Only the Neuropathy Symptom Score showed a trend that did not reach significance. The treatment did raise thiamine levels, so it was absorbed and doing something biochemically. It just did not translate into benefit. It was well tolerated, and that is the most positive thing that can honestly be said about it.

**Acetyl-L-carnitine, and this one is the most important sentence in the guide if you have chemotherapy-related neuropathy.** Acetyl-L-carnitine is widely sold for nerve support, including specifically to people going through chemotherapy. In SWOG S0715, a randomized, double-blind, multicenter trial of 409 women receiving taxane-based chemotherapy for breast cancer, acetyl-L-carnitine at 1000 mg three times daily for 24 weeks **made neuropathy worse, not better.**<sup>9</sup> The trial's own words are that it "was unexpectedly found to increase CIPN." The difference was statistically significant, and it was still measurable at two years, long after the supplement had stopped. At 104 weeks, 39.5% of the carnitine group versus 34.4% of the placebo group reported a meaningful worsening from baseline.

There is no version of that result that supports selling acetyl-L-carnitine to this audience. If your neuropathy is chemotherapy-related, duloxetine remains the only pharmacological treatment recommended by ASCO,<sup>10</sup> and that is a conversation with your oncology team.

**High-dose vitamin B6 as a treatment.** Covered above. There is no evidence for it as a standalone treatment, and a real risk of causing the symptoms you are treating.

**Proprietary blends with undisclosed amounts.** If a label says "nerve support complex 850 mg" without breaking out what is in it, you cannot total your B6, you cannot compare it to a studied dose, and you cannot tell your prescriber what you are taking. That is disqualifying on safety grounds alone.

**Anything sold as reversing or curing neuropathy.** No supplement has demonstrated this. ALA, the best-evidenced product in the category, missed its primary endpoint in the one trial designed to test disease modification over years.<sup>6</sup>

**Products I could not find honest evidence for.** Several ingredients common in nerve formulas have mechanistic rationale and no meaningful trial data in this condition. Their absence here is not an oversight.

# Running a fair 12-week trial

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The most common way a guide like this gets wasted is changing six things at once. Then something improves, or does not, and you have learned nothing you can act on.

**One change at a time.** Yes, it is slower. It is also the only way to end up knowing something.

**Write down where you are starting.** Before you take anything, record what your symptoms are like now: how bad at their worst, how often, what they stop you doing, and how you slept. Memory is unreliable about gradual change in both directions, and this is what the free tracker is for.

**Give it long enough, but not forever.** Symptom-level effects in the ALA trials appeared over about five weeks.<sup>3</sup> Nutrient correction takes longer, because you are rebuilding rather than modulating. Twelve weeks is a reasonable outer bound for deciding whether something earned its place.

**Decide in advance what success looks like.** “A bit better maybe” is not a result. Half as much burning at night, or sleeping through instead of waking twice, is a result.

**If it did nothing, stop taking it.** This is the step people skip. An unhelpful supplement that stays in the routine forever is how someone ends up on nine things, spending real money, with no idea which ones matter.

**Track it:** [rootcauseunlocked.com/tracker?utm\\_source=guide&utm\\_medium=ebook&utm\\_campaign=neuropathy-supplement](https://rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=neuropathy-supplement)

Turn on recovery tracking if nerve symptoms limit what you can do day to day. It is free, it works anywhere, and your entries stay on your own device.

# Realistic expectations

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Here is what the numbers in this guide actually mean for a person.

In the best short-term trial, 62% of people on 600 mg of ALA achieved at least a 50% reduction in symptom score, against 26% on placebo.<sup>3</sup> That is a real difference and it is also not everyone. Roughly a quarter of people improved on placebo alone, which tells you something honest about how variable these symptoms are week to week.

Vitamin D reduced short-term pain by roughly 11 points on common pain scales in painful diabetic neuropathy, in four small trials, with no long-term data at all.<sup>8</sup>

B vitamins improved some measures and not others, and the improvement in examination scores did not reach the threshold considered clinically meaningful.<sup>7</sup>

None of these is a cure, none was tested for years with a positive result, and the one trial that did run for years missed its primary endpoint.<sup>6</sup> What they add up to is a reasonable chance of meaningfully less burning and tingling, on top of addressing the cause, not instead of it.

If your neuropathy has a driver that can be found, finding it will do more than anything on this list.

# When to seek professional care

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**Immediately, without waiting on anything here:** rapidly progressing weakness, symptoms spreading over days rather than months, a fall related to numbness or balance loss, or any difficulty with breathing or swallowing alongside nerve symptoms.

**Promptly:** new or worsening symptoms after starting a supplement, any of the serotonin syndrome symptoms listed earlier, or nerve symptoms that began after starting a new medication.

**At your next appointment:** if you take metformin or a proton pump inhibitor and have never had B12 checked with methylmalonic acid, or if you are taking more than 100 mg of B6 daily across everything.

# Dr. Seay's Favorite Support Products

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Everything above was written without reference to any product line, and the order of this guide reflects the strength of the evidence, not what is easy to buy.

**Look for third-party testing first.** NSF Certified for Sport and USP Verified both mean an independent lab confirmed the product contains what the label says and is free of specified contaminants. Supplements are regulated as food rather than as drugs, so label accuracy is not automatic. This matters more than brand.

**Then check the four things this guide has given you:** that the amount matches what was studied, that the form matches what was studied, that the full ingredient list is disclosed rather than hidden in a proprietary blend, and that the total B6 across everything you take stays under 100 mg per day.

If you would like products already screened on those criteria, my dispensary is here:

[us.fullscript.com/welcome/drseay](https://us.fullscript.com/welcome/drseay)

**Disclosure:** Root Health earns a portion of sales made through that link. Nothing in this guide was included, excluded, ranked, or dosed based on what it earns. Benfotiamine sells well and is in this guide's do-not-bother section because the twelve-month trial was negative. You can buy any of this anywhere, and if you already have a source you trust, use it.



# Working with Root Health

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Consultations are available to people located in North Carolina, where I am licensed. If you are elsewhere, take this guide's medication table and the tier 3 question list to your own clinician. They work just as well in their office as in mine.

**Not ready for a conversation yet? Start tracking:** [rootcauseunlocked.com/tracker?](https://rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=neuropathy-supplement)  
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For educational purposes only. This guide does not constitute medical advice, diagnosis, or treatment, and reading it does not establish a patient-provider relationship.

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